

Celleccts: a software to quantify cell expansion and motion

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Software

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Summary

Celleccts is a user-friendly and open-source software for automated quantification of biological growth, motion, and morphology from 2D image data and time-lapse sequences (2D + t), acquired under a wide range of experimental conditions and biological systems (from fungal colonies to unicellular branching networks). The software is available as a stand-alone version, featuring a graphical interface that supports interactive parameter tuning, visualization, validation, and batch processing. The analysis pipeline can be extended and customized using a dedicated Python API.

The typical inputs and outputs are as follows. Celleccts is designed to process grayscale or color images originating from standard microscopy, macroscopic imaging setups, or camera-based platforms. The software supports single or multiple organisms growing or moving in one or several arenas and can analyze multiple folders sequentially. All quantitative results (area, circularity, orientation axes, centroid trajectories, oscillations, network topology) are exported as standardized .csv files suitable for downstream statistical analysis, ensuring reproducibility and integration into existing workflows.

Statement of need

Modern imaging generates high-resolution biological datasets across scales, yet automated analysis remains challenging for non-experts, necessitating accessible tools.

Celleccts is suited to biological systems exhibiting continuous growth, deformation, or collective motion, such as fungal colonies (Figure 1a-d), HeLa cells (Figure 1e-h), and slime molds (Figure 1i-n). By contrast, most existing tools target single species (mainly yeast or bacteria) and fail to generalize to heterogeneous morphologies such as branching slime mold networks or collective cellular movement during proliferation.

Open source alternatives often lack graphical user interfaces (GUIs) and robust automation under variable lighting/contrast conditions, while commercial platforms often require preprocessing or post-analysis using additional software, compromising reproducibility.

By combining dynamic segmentation algorithms with a modular pipeline (see Software Design), Celleccts supports both single-specimen analysis and high-throughput multi-arena experiments, outputting standardized metrics directly usable in downstream statistical workflows. While enabling reproducible studies across diverse biological models, this automated quantification reduces observer bias.

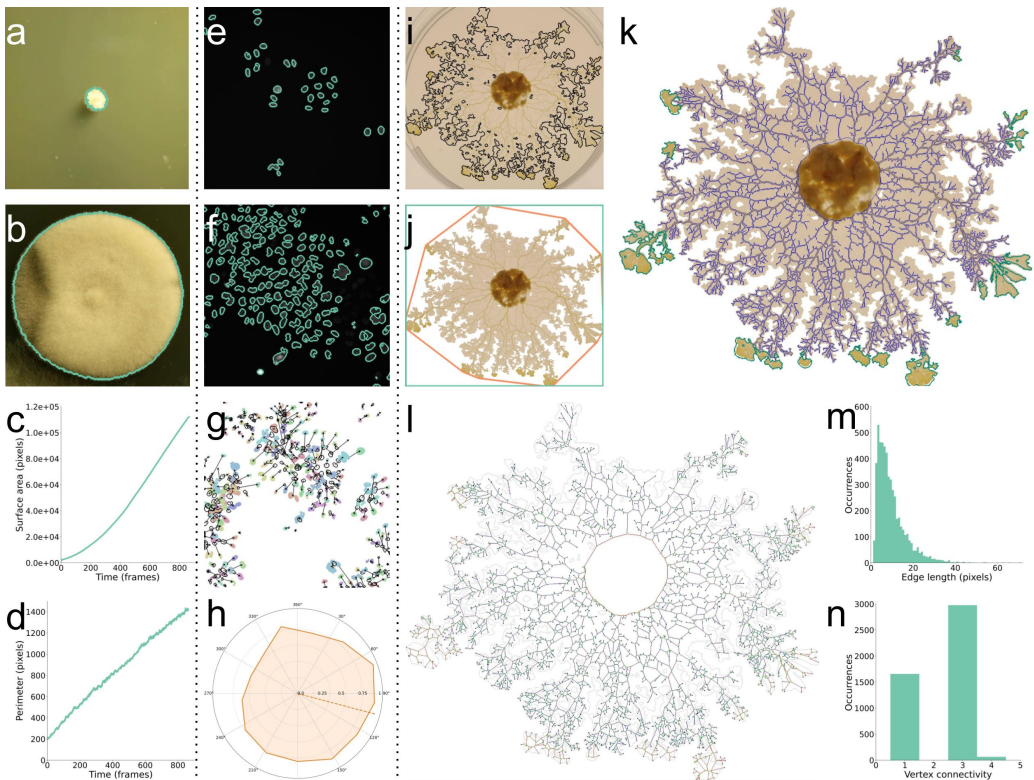


Figure 1: Cellular dynamics and morphologies across systems. **a,b**) Fungal growth (unknown sp.) from initial (**a**) to final (**b**) stages with green segmentation contours, from (Peñil Cobo et al., 2018); **c,d**) Corresponding area and perimeter curves over time (**c**: area, **d**: perimeter). **e,f**) Tracking of HeLa “Kyoto” cells marked with mCherry-H2B, from (Guet, 2022) showing initial (**e**) and final (**f**) images with segmentation contours. **g**) Motion vectors (arrows) of the 250 most mobile cells among 1319 detected (black contour: original positions, colored-filled patches: final locations). **h**) Spider plot representing HeLa cell movement directions. **i,j**) *Physarum polycephalum* morphology after 16:40 hours of exploration (this study). (**h**: cell segmentation, **i**: convex hull (orange) and bounding rectangle (green)). **k,l**) Network segmentation (**j**: blue network; turquoise pseudopods) and graph reconstruction (**k**: edges colored by width from blue to red, green branching vertices, black tips, yellow food vertices). **m,n**) *Physarum* connectivity metrics: edge lengths (**l**) and vertex degrees (**m**). Panels **a–d** (fungus), **e–h** (HeLa), **i–n** (*Physarum*).

State of the field

Collects fills three major gaps in existing tools:

Limitation	Existing solution	Collects innovation
Taxon-specific design	Bacteria (Ernebjerg & Kishony, 2012), Yeast (Falconnet et al., 2011)	Universal pipeline adaptable to fungi, animals, and slime molds via tunable segmentation parameters
Interface barriers	Manual scripting (Pandey et al., 2021), Commercial solutions (ScanLag (Levin-Reisman et al., 2014) or ColTapp (Bär et al., 2021))	GUI + API dual architecture for both non-programmers and developers
Output limitations	Needs post-analysis manual processing with tools like ImageJ/Fiji (Schneider et al., 2012)	Precomputed shape descriptors, network topology data (vertex degrees, edge widths), and CSV exports

42 While commercial tools prioritize ease of use over customization, open-source platforms often
43 require manual scripting. Collects bridges this divide by combining automation with a validation
44 tool for result refinement, enabling robust and accessible analysis of growth dynamics across
45 biological systems and fostering reproducibility and cross-disciplinary research.

46 Software design

47 The software is organized around a layered architecture centered on a global controller
48 (*ProgramOrganizer*), which maintains experiment state, configuration parameters, and
49 processing context. This controller can be driven either through the graphical user interface or
50 programmatically.

51 The graphical interface follows a sequential workflow implemented via a stacked widget
52 (*CollectsMainWidget*), exposing successive stages for data loading (*FirstWindow*),
53 segmentation and arena definition (*ImageAnalysisWindow*), and time-series execution
54 (*VideoAnalysisWindow*). This structure mirrors the experimental pipeline and limits user
55 interaction to valid analysis states while allowing iterative refinement at each step. To balance
56 accessibility with flexibility, novice users benefit from an automated parameter search during
57 initial setup (in *ImageAnalysisWindow*), while experienced ones can bypass default algorithms
58 in advanced mode to directly customize image processing settings.

59 Static and temporal analyses are separated through two dedicated classes: *OneImageAnalysis*,
60 responsible for preprocessing tasks such as greyscale conversion, filtering and background
61 subtraction, and *MotionAnalysis*, which performs segmentation, post-processing, video-based
62 measurements and temporal feature extraction. This separation allows computationally intensive
63 motion analysis to build upon validated segmentation results.

64 Collects targets diverse biological datasets (e.g., Fungi, HeLa cells, Myxamoebae) acquired under
65 variable imaging conditions. Rather than relying on a single segmentation model, the image
66 analysis layer provides automated and configurable pipelines combining multiple color-space
67 representations, image filtering, K-means clustering and threshold-based methods. Geometric
68 descriptors (area, perimeter, circularity) are encapsulated in the *ShapeDescriptors* class, while
69 additional modules support graph-based dynamics, oscillatory behavior, and morphological
70 operations.

71 To maintain interactivity during heavy computation, Collects combines Qt-based threading
72 for GUI responsiveness with multiprocessing for video analysis. Memory usage is explicitly
73 managed through sequential image processing and controlled data release.

74 Research impact statement

75 Related work

76 Collects' lineage traces directly to (Vogel et al., 2015)'s MATLAB implementation, with an
77 early iteration already employed by (Boussard et al., 2019). While the final version has not
78 yet enabled new studies, it has been developed to address specific research questions from
79 (Boussard et al., 2021) and from an ongoing ANR project (ANR-24-CE45-3362, PI Claire
80 David).

81 Validation

82 The software's robustness is demonstrated through specimen and background accuracies (Figure
83 2) in diverse contexts. First, manual segmentation of Figure 1 examples provided ground truth
84 for canonical cases: single fungus on color-varying background Figure 2a-c, multi-specimen
85 tracking Figure 2d-f with microscopy data, and network extraction Figure 2g-i. Second, Collects
86 was tested on highly heterogeneous cells where manual distinction is infeasible. Accuracy was

87 estimated via error annotations (Figure 2j) comparing original images to results, achieving
88 >97% accuracy across five challenging conditions (Figure 2k: high contrast + optimal setup,
89 heterogeneous colors, low contrast + desiccation, low contrast + optimal, very low resolution)
90 through iterative parameter refinement in the GUI.

91 Applications span fungal growth tracking (Figure 1a-d, (Peñil Cobo et al., 2018)), HeLa cells
92 motion (Figure 1e-h, (Guet, 2022)), and network analysis of *P. polycephalum* (Figure 1i-n,
93 this study), highlighting adaptability to diverse biological systems and imaging setups.

DRAFT

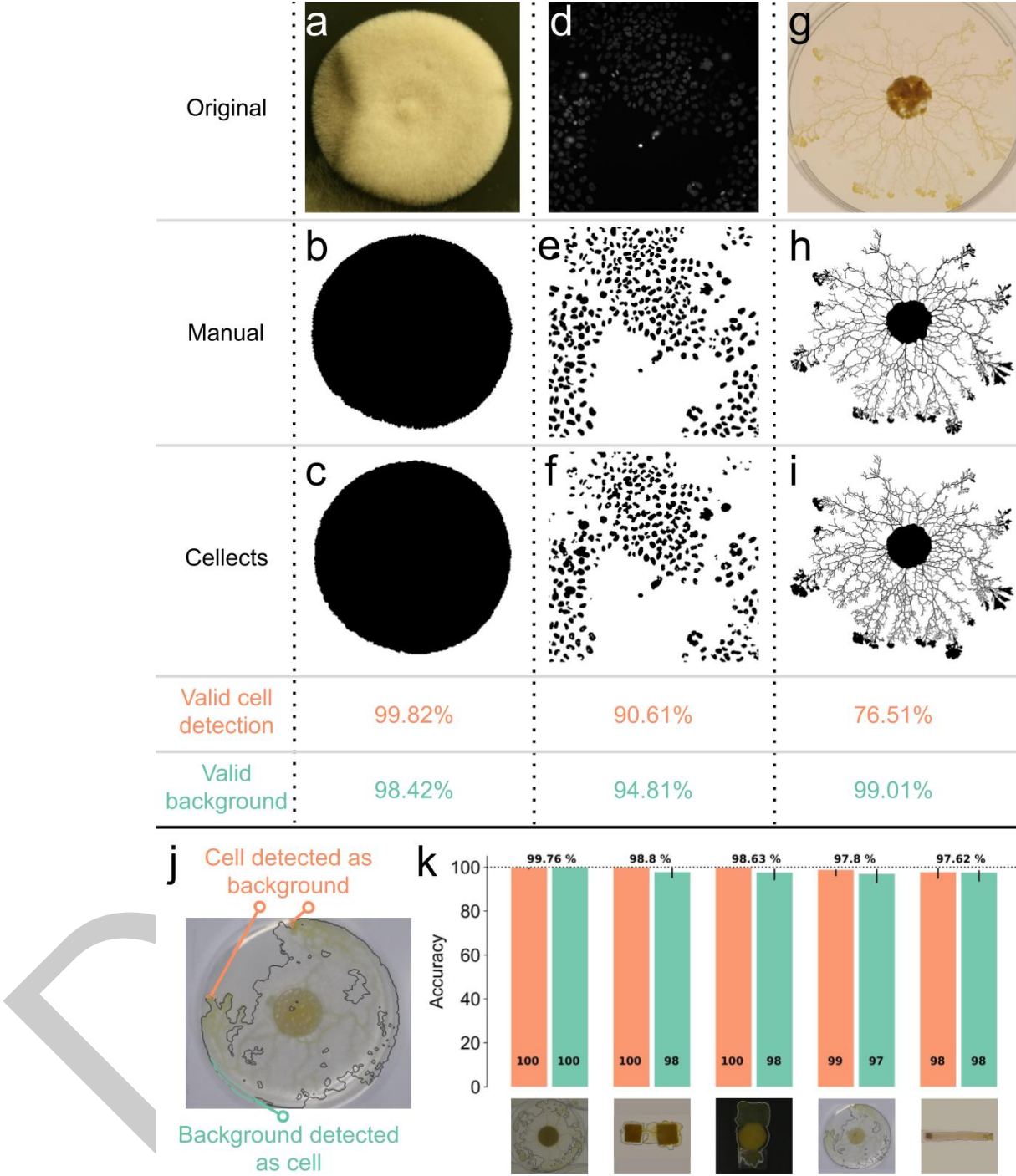


Figure 2: Validation of Collects across five experimental conditions. **a,b,c**) Segmentation accuracy of the fungi (Peñil Cobo et al., 2018) against ground truth: original image (a) was used to create a mask manually (b) and using Collects (c). The valid cell detection is the percentage of pixels accurately labelled as cells by Collects. The valid background is the percentage of pixels accurately labelled as background by Collects. **d,e,f**) Segmentation accuracy of the Hela “Kyoto” cells (Guiet, 2022) against ground truth. **g,h,i**) Segmentation accuracy of the *P. polycephalum* network against ground truth (this study). **j**) Image of a *P. polycephalum* plasmodia, showing the two types of errors detected during validation: background pixels classified as cell (orange), cell pixels classified as background (green). **k**) A posteriori accuracy in 5 experimental conditions (shown below the bars): high contrast with optimal setup, high contrast with heterogeneous colors, low contrast with a setup prone to desiccation, low contrast with optimal setup, low resolution. Orange: Proportion of cell pixels correctly identified as cell. Green: Proportion of background pixels correctly identified as background. Error bars show the 95% confidence interval. Percentages on top show the average of both bars.

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Author contribution

Conceptualization, A.B., P.A., A.D., A.P.E.; methodology, A.B., M.P., A.P.E, P.A.; software, A.B.; investigation, A.B., A.D.; resources, A.B., P.A., A.D., A.P.E.; writing – original draft preparation, A.B.; writing – review and editing, A.B., A.P.E., M.P., A.D.; visualization, A.B.; supervision, P.A., A.D., A.P.E.; project administration, A.B.; funding acquisition, A.D., A.P.E.

AI usage disclosure

The generative model 'Devstral-Small-2507' was used to generate initial drafts of function docstrings and propose unit test templates. All AI-generated content underwent manual verification to ensure alignment with function usages in the context of real datasets.

Data availability

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Data and code availability

The Windows and macOS versions are accessible via the following link: <https://github.com/Aurele-B/Collects/releases>.

The software documentation is available at <https://aurele-b.github.io/Collects> and its source code can be found at <https://github.com/Aurele-B/Collects>.

To access the data and replication code, refer to: <https://datadryad.org/stash/share/nCvWIZoZ8-Wnxm0CjnPbbznUPw90RYdo1YVJEQkLIY>

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